

YELETS, Russian Federation

WMO: 279280

Lat: 52.633N

Lon: 38.517E

Elev: 549

StdP: 14.41

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-6.1	-16.7	2.3	-11.4	-11.2	3.1	-5.4	15.5	24.5	14.1	23.3	3.1	30	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	88.9	67.3	85.5	66.4	82.2	65.1	70.1	82.5	68.6	80.5	67.3	78.4	5.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.4	74.6	64.8	93.7	73.1	63.4	89.3	72.0	34.4	82.8	33.1	80.7	32.0	78.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.9	12.1	10.9	DB	-16.2	92.7	5.9	4.0	-20.5	95.6	-23.9	98.0	-27.2	100.2	-31.5	103.2	
			WB	-16.6	72.6	5.8	1.7	-20.8	73.8	-24.2	74.8	-27.4	75.8	-31.7	77.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.3	19.2	19.8	29.6	46.3	58.9	64.9	69.3	66.5	56.0	44.1	32.0	23.3
	DBStd	20.27	12.24	12.31	8.88	8.51	8.49	6.83	6.13	6.83	7.62	8.06	9.56	11.37
	HDD50	4239	956	845	633	171	20	1	0	0	28	218	540	827
	HDD65	7957	1421	1265	1098	560	226	86	26	63	282	648	990	1292
	CDD50	2161	0	0	0	60	297	448	599	512	209	35	1	0
	CDD65	404	0	0	0	0	38	83	160	110	13	0	0	0
	CDH74	3847	0	0	0	11	397	733	1427	1126	153	0	0	0
	CDH80	1227	0	0	0	1	93	204	496	410	23	0	0	0
Wind	WSAvg	5.1	5.7	5.9	5.9	5.6	4.8	4.5	3.9	4.0	4.1	5.1	5.6	6.0
Precipitation	PrecAvg	22.8	1.7	1.4	1.3	1.6	1.7	2.6	2.9	2.2	2.0	2.1	1.7	1.5
	PrecMax	32.6	4.0	3.0	2.5	3.2	4.2	5.6	4.9	7.3	5.1	5.5	4.3	4.4
	PrecMin	13.5	0.6	0.3	0.3	0.1	0.1	0.4	0.3	0.3	0.1	0.3	0.1	0.4
	PrecStd	4.1	0.7	0.6	0.7	0.9	0.9	1.3	1.2	1.7	1.3	1.2	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	38.9	57.2	75.7	86.8	90.0	94.5	96.3	83.2	69.7	54.8	45.9
		MCWB	37.8	36.2	46.0	55.4	64.6	65.6	68.2	67.1	61.2	56.9	51.6	43.9
	2%	DB	36.6	36.9	48.7	69.8	82.7	85.5	88.9	88.7	78.7	64.0	50.3	41.6
		MCWB	35.0	34.9	41.4	53.8	62.4	66.0	68.3	67.3	60.4	55.3	48.2	40.1
	5%	DB	34.9	35.5	44.1	65.6	78.6	81.6	85.6	84.3	74.0	59.3	47.1	38.4
		MCWB	33.6	33.7	38.6	51.8	61.4	65.4	67.8	66.1	59.4	53.1	45.0	37.2
	10%	DB	33.6	34.2	40.3	61.3	74.4	78.1	81.8	80.1	69.3	55.5	43.9	35.5
		MCWB	32.5	32.9	36.7	49.9	59.5	63.9	66.4	64.1	58.1	51.0	42.1	34.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	37.3	46.8	57.9	67.3	70.7	72.4	71.7	64.8	59.8	51.7	44.6
		MCDB	39.2	38.4	54.5	68.2	81.3	81.0	86.1	84.6	75.2	65.8	54.5	45.9
	2%	WB	35.3	35.5	42.6	55.2	64.7	68.7	70.6	69.4	62.7	56.8	48.5	40.3
		MCDB	36.1	36.4	47.6	65.9	78.2	79.7	83.5	82.2	73.3	61.9	50.2	41.4
	5%	WB	33.8	34.1	39.5	53.1	62.7	67.1	69.2	67.5	60.9	54.0	45.2	37.3
		MCDB	34.6	35.0	43.2	63.3	74.5	77.7	81.2	80.5	69.6	58.6	46.8	38.3
	10%	WB	32.6	33.1	36.9	50.9	60.8	65.6	67.9	65.8	58.9	51.4	42.3	34.5
		MCDB	33.6	34.2	39.7	59.9	71.9	75.4	78.8	77.3	67.8	55.0	43.7	35.2
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.7	11.9	17.3	19.3	17.8	18.4	20.1	17.3	12.4	8.2	8.3
		MCDBR	7.0	7.4	16.7	23.9	24.4	23.4	23.6	26.3	25.8	19.4	11.6	7.2
		MCWBR	6.6	6.7	10.4	11.4	9.6	8.7	7.8	9.1	10.8	11.4	9.7	6.6
	5% WB	MCDBR	6.7	7.1	15.1	21.0	21.8	19.6	20.3	22.5	20.5	16.4	10.2	6.8
		MCWBR	6.8	6.9	10.1	10.9	9.6	8.2	7.9	8.9	10.3	10.9	9.2	6.4
Clear-Sky Solar Irradiance	taub	0.298	0.335	0.359	0.435	0.395	0.386	0.420	0.438	0.388	0.360	0.334	0.302	
	taud	2.117	2.027	2.084	2.084	2.308	2.348	2.268	2.203	2.321	2.342	2.258	2.152	
	Ebn at Noon	209	234	258	252	272	275	262	249	249	230	197	184	
	Edh at Noon	23	34	40	46	38	37	40	40	32	25	21	19	
All-Sky Solar Radiation	RadAvg	282	604	1065	1332	1806	1862	1827	1554	1024	522	223	155	
	RadStd	66	70	167	115	142	184	166	145	128	76	45	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.55	N/A	N/A	N/A	N/A	N/A	-292	-457	+245	N/A
(47)	Regional (9 neighbors)	+1.44	N/A	N/A	N/A	N/A	N/A	-254	-374	+212	+76

Nomenclature: See separate page